Samba Gangineni

samba-gangineni   samba.sr.gangineni@gmail.com   857 425 9396   samba-gangineni

Atlanta, GA

Profile

Senior Software Engineer with 7+ years of experience architecting and delivering scalable backend systems, distributed data pipelines, and production-grade AI applications. Strong track record building microservices (Java Spring Boot, Python FastAPI, Node.js), ETL/training pipelines (Apache Spark/Scala), and AI-powered customer experiences including LLM agents and evaluation pipelines with observability, reliability, and cost controls across cloud environments (AWS, GCP).

Skills

- **Programming:** Java, Kotlin, Python (FastAPI, Flask), JavaScript/Node.js, Go, SQL, Bash, HTML, CSS, HCL
- **Cloud & DevOps:** AWS, GCP, Docker, Kubernetes, Helm, Terraform, Pulumi, Git, Jenkins, CI/CD, Redis
- **Data & AI Systems:** Apache Spark, Scala, Temporal, RabbitMQ, Elasticsearch, Vector Search (e.g., FAISS/Weaviate), LangChain, LangGraph, Google ADK, LangSmith (observability/evals)
- **ML/LLM:** Evaluation pipelines, prompt tuning, RAG, agent/tool orchestration, TensorFlow/PyTorch
- **Databases:** MongoDB, MSSQL Server

Experience

The Home Depot — *Jun 2025 – Present — Atlanta, GA*
Senior Software Engineer

- Built and enhanced the **Magic Apron Assistant** (LLM chatbot) using agent orchestration frameworks (e.g., Google ADK, LangChain/LangGraph), improving response quality through structured tool calling and workflow routing.
- Designed the **Planner Assistant** architecture to unify Live Chat and Magic Apron, including session handling, conversation memory, and scalable agent routing using ADK-based patterns.
- Own and maintain **RexAPI** for model serving and recommendation integrations, supporting batch/online workflows and production reliability (APIs, validation, error handling, and operational monitoring).
- Contributed to the **Magic Forge** evaluation pipeline by building and running automated eval suites with metrics and gating (groundedness/faithfulness, hallucination rate, tool-call success rate, response formatting validity, latency, and cost), and enforcing regression thresholds to block prompt/agent changes that degraded quality.
- Implemented **feedback loops** for LLM features, including evaluation dataset curation (goldens) for new capabilities, human review workflows, and capturing user feedback to drive iteration; partnered on automated failure triage workflows to expand regression coverage.
- Implemented end-to-end **LangSmith** observability for Magic Apron (traces, datasets, and run comparisons) to support debugging, evaluation analysis, and release regression tracking.
- Implemented **rate limiting and backpressure** controls to protect downstream model/tool dependencies and improve stability under load.
- Collaborated with Google on a specialized **Garden Agent** solution for garden-related customer experiences, leveraging Magic Apron capabilities and integrating domain-specific tools and flows.
- Helped overhaul **recommendations MLOps pipelines** to enable data science teams to iterate faster, including training workflow improvements, automated metric publishing, and pipeline validation for reliable

releases.

Availity *Feb 2020 – Jun 2025 — Atlanta, GA*
Software Engineer IV

- Led a redesign of a legacy application by integrating Apache Spark pipelines and Scala, enhancing performance & scalability and reducing cost by 90% to support multi-tenant workloads.
- Developed an internal Confluence chatbot for knowledge management — implemented REST endpoints using Python FastAPI, integrated LLMs with Retrieval-Augmented Generation (RAG), and leveraged vector search for efficient retrieval and relevance ranking.
- Developed an AI agent to generate documentation from function signatures/metadata by constructing prompts and using Amazon Q integrated with LangChain to produce human-readable documentation.
- Designed and implemented a stateless product pipeline with event-driven architecture using Temporal, reducing tight coupling and improving reliability.
- Transitioned infrastructure code and customer environments from Pulumi to Terraform, improving provisioning repeatability and deployment stability.
- Achieved a 5x performance improvement in data deduplication by redesigning the architecture with Kotlin coroutines to minimize database and RabbitMQ load.
- Streamlined data ingestion using Temporal and managed CI/CD deployments into EKS with Terraform.
- Spearheaded MongoDB sharding and large-scale dataset regeneration, supporting high-volume processing workloads.
- Managed large-scale ingestion pipelines using Spark, AWS Lambda, EMR, and Kubernetes for multi-tenant processing at scale.
- Helped build a **batch clinical inference model** to detect gaps in structured datasets using association mining, including feature preparation, batch scoring, and evaluation of output quality.
- Used **TensorFlow** and **PyTorch** professionally for model experimentation and evaluation (e.g., generative modeling and text generation prototypes), including training runs, tuning, and result analysis.
- Automated terminology updates with Python and SQL Server stored procedures to support normalization and downstream analytics.

Boston University *May 2018 – Dec 2019 — Boston, MA*
Graduate Research Assistant / Teaching Assistant (Big Data Analytics with Spark)

- Conducted graduate research on large-scale datasets using Python-based pipelines; produced reproducible experiments, metrics, and documentation for publication-quality results.
- Supported Spark labs and office hours as TA, debugging assignments and teaching Spark concepts (transformations/actions, shuffles/partitioning, Spark SQL).

Diameter Health *Jun 2019 – Aug 2019 — Boston, MA*
Informatics and Engineering Intern

- Developed a transpiler in Golang to convert Clinical Quality Language into JavaScript, automating boilerplate code generation.
- Designed and implemented a synthetic data generator for clinical HL7 messages in JavaScript, accelerating testing and development cycles.

Research

Synthetic Health Data Generation *Jan 2026*
TE-CTABGAN+: A Unified GAN Framework for Generating Synthetic Electronic Health Records

- Researched methods to generate synthetic healthcare data using MIMIC-IV and Synthea sources; compared GAN and Diffusion approaches and incorporated temporal modeling to improve realism.

- Publication: https://doi.org/10.1007/978-3-032-15632-7_20

ACM DEBS *Jun 2019*
Distributed and Event Based Systems

- Architected a real-time object detection system for LiDAR data using clustering techniques and convolutional neural networks.
- Publication: https://doi.org/10.1145/3328905.3330297

Education

Boston University *MSc in Computer Info Systems* *Grad. Jan 2020*

- GPA: 3.86/4.0
- **Coursework:** Big Data Analytics with Spark, Artificial Intelligence, DataMining & Business Intelligence

Awards

Professional Awards	Jan 2026	Executive Home Award (The Home Depot) – Magic Apron Outdoor Assistant
	Mar 2026	Orange Spotlight Award (The Home Depot) – Magic Apron Outdoor Assistant
Academic/Hackathon Awards	2019	ACM DEBS Grand Challenge Winner (Object Recognition)
	2018	1st Place in BigRedHacks (ResQu - An SOS Application)
	2018	3rd Place in Att Hacks (Imound - An App for the Visually Challenged)